# Pro Football Sports Science Research Update

## April 2024

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**Assessing isometric kicking force and post-match responses using the Kicker test. Journal of Sports Sciences, 40(11): 1275-1281, 2022.**

**Cannabis containing THC impairs 20-min cycling time trial performance irrespective of the method of inhalation. Journal of Applied Physiology, 136:583-591, 2024.**

**Circadian rhythm in sportspersons and athletic performance: A mini-review. Chronobiology International: The Journal of Biological and Medical Rhythm Research, 41(2): 137-181, 2024.**

**Current evidence on common dietary supplements for sleep quality. American Journal of Lifestyle Medicine, Ahead of Print: 1-5, 2024.**

**Effects of mindfulness-based interventions on promoting athletic performance and related factors among athletes: A systematic review and meta-analysis of randomized controlled trial. International Journal of Environmental Research and Public Health, 20:2038, 2023.**

**Exposures to elevated core temperatures during football training: The impact on autonomic nervous system recovery and function. Sports, 12: 8, 2024.**

**Football practices in hot environments impact subsequent days' hydration. Journal of Strength and Conditioning Research, 38(1):90-96, 2024.**

**Hamstring Injuries: A paradigm for return to play. Seminars in Musculoskeletal Radiology, 28:119-129, 2024.**

**Heel Pain – Plantar Fasciitis: Revision 2023. Journal of Orthopaedic and Sports Physical Therapy, 53(12): CPG1-CPG39, 2023.**

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**Low Ankle-GO Score while returning to sport after lateral ankle sprain leads to a 9-fold increased risk of recurrence: A two-year prospective cohort study. Sports Medicine – Open, 10:23, 2024.**

**Many pieces to the puzzle: A new holistic workload approach to designing practice in sports. Sports Medicine – Open, 9:38, 2023.**

**Mechanical and metabolic power in accelerated running – Part 2: Team sports. European Journal of Applied Physiology, 124:417-431, 2024.**

**More than metrics: The role of socio-environmental factors in determining the success of athlete monitoring. Journal of Sports Sciences, Ahead of Print: 1-11, 2024.**

**NCAA football players are at higher risk of upper extremity injury after first-time concussion. The Physician and Sportsmedicine, Ahead of Print: 1-16, 2024.**

**On the value of a premium college football player: Evaluating the literature. Journal of Sports Economics, Ahead of Print: 1-35, 2024.**

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**Patellar tendon load progression during rehabilitation exercises: Implications for the treatment of patellar tendon injuries. Medicine and Science in Sports and Exercise, 56(3):545-552, 2024.**

**Performance-enhancing substances in sport: A scientometric review of 75 years of research. Drug Testing and Analysis, Ahead of Print: 1-12, 2024.**

**A profile of isometric cervical strength in elite professional male rugby players. Journal of Orthopaedic and Sports Physical Therapy, 54(3):1-8, 2024.**

**Recovery management in sport: Overview and outcomes of a nine-year multicenter research program. International Journal of Sports Science & Coaching, Ahead of Print: 1-11, 2024.**

**Repetitive head injuries in German American football players do not change blood-based biomarker candidates for CTE during a single season. Neurological Research and Practice, 5:13, 2024.**

**The 11 evidence-informed and inferred principles of microcycle periodization in elite football. Sport Performance & Science Reports, Feb: 218, 2024.**

**The differentiation of the competitive athlete with physiologic cardiac remodeling from the athlete with cardiomyopathy. Current Problems in Cardiology, Ahead of Print: 1-47, 2024.**

**The impact of different warm-up strategies on acceleration and deceleration demands in highly trained soccer players. European Journal Sport Science, 24:88-96, 2024.**

**The mental health of elite-level coaches: A systematic scoping review. Sports Medicine – Open, 10:16, 2024.**

**The sleep parameters of Olympic athletes: Characteristics and assessment instruments. International Journal of Sports Medicine, Ahead of Print: 1-9, 2024.**

**The top 100 most impactful articles on the Achilles tendon according to altmetric attention score and number of citations. The Orthopaedic Journal of Sports Medicine, 12(3): 1-5, 2024.**

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# Pro Football Sports Science Research Update

**Validity of global positioning system technology to measure maximum velocity sprinting in elite sprinters. Journal of Strength & Conditioning, 37(12):2438-2442, 2023.**

**Wearable and ingestible technology to evaluate and prevent exertional heat illness: A narrative review. PM & R: The Journal of Injury, Function, and Rehabilitation, Ahead of Print: 1-6, 2024.**

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# Pro Football Sports Science Research Update

## *Anterior pelvic tilt increases hamstring strain and is a key factor to target for injury prevention and rehabilitation. Knee Surgery, Sports Traumatology, Arthroscopy, 32:573-582, 2024.*

**Purpose:** Hamstring muscle strain injury is very common in sports involving high-speed running. Hamstring muscles originate from the ischial tuberosity and thus pelvic position may influence hamstring strain during different sports movements like sprinting, but this has only been evaluated by indirect methods. This study tested the hypothesis that a change in anterior pelvic tilt causes elongation of the overall hamstring complex and disproportionately elongates proximal relative to distal muscle regions.

**Methods:** Seven fresh-frozen specimens (full lower limb with pelvis and lumbar spine) were used for this in vitro study. Specimens were dissected to enable visualization of the hamstring muscles and then fixed into a custom-made testing bench that allowed controlled movement of the pelvis over a fixed femur and tibia. Nine markers were inserted into the hamstring muscles to allow intra- and intermuscle difference measurements. Then, six different anterior pelvic angles were used to measure the difference in hamstring muscle lengthening through a three-dimensional reconstruction system based on stereoscopic machine vision technology.

**Results:** An increase in anterior pelvic tilt produced a significant non-uniform increase in tissue elongation in all regions of the three hamstring muscles (semitendinosus, semimembranosus [SMB] and biceps femoris long head), which was greater in the proximal (>1 cm every 5°) compared to the distal region (≈0.4 cm every 5°). At the proximal hamstring region, SMB showed significantly greater length changes compared to conjoint tendons with nonstatistically significant elongation differences between muscles at the distal region.

**Conclusion:** Considering the results of the study, the pelvis segment will likely play a fundamental role as a strain regulator of hamstring muscles. These results will have an impact on injury rehabilitation and prevention processes of hamstring injuries, as well as optimize future musculoskeletal models and avoid potential underestimation of the hamstring muscle-tendon complex lengthening during high-speed running.

**Level of evidence:** N/A.

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## *Assessing isometric kicking force and post-match responses using the Kicker test. Journal of Sports Sciences, 40(11): 1275-1281, 2022.*

**Abstract -** This study examined the inter-session reliability of force output from a novel isometric strength assessment protocol (the Kicker); and its suitability to monitor soccer player's combined hip flexion and knee extension force capacity over 72-h post-competitive matches. Reliability (Part-A) testing was completed over three sessions on 20 individuals participating in various sports at a recreational level or higher. Post-match strength response (Part-B) data were collected for 72-h after a game (24-h (+24), 48-h (+48) and 72-h (+72) post-match) in 17 male academy soccer players. After familiarization, Kicker force for each limb showed high inter-session reliability (ICC >0.95; typical error <14 N, CV <6%); minimum detectable change at a 95% confidence interval <40 N). Across the 72-h post-match period, Kicker force for each limb was suppressed compared to baseline (force loss range = -5.8% to -12.5%; effect sizes range = -0.26 to -0.43) at all time points. The Kicker assessment protocol measures combined isometric hip flexor and knee extensor force capacity with high inter-session reliability. The proof of concept that the protocol can be used as a monitoring tool was evidenced by sustained suppression of baseline force capacity in both kicking limbs for 72-h post soccer matches.

## *Cannabis containing THC impairs 20-min cycling time trial performance irrespective of the method of inhalation. Journal of Applied Physiology, 136:583-591, 2024.*

**Abstract -** Herein, we examine the human exercise response following cannabis inhalation, taking into consideration varied cannabinoid concentrations and different inhalation methods. A semi-randomized crossover study design was used, with measures of perceived exertion and physiological responses to submaximal and maximal exercise. Participants (n = 14, 9 males 5 females) completed exercise after 1) smoking Δ-9-tetrahydrocannabinol (THC)-predominant cannabis (S-THC), 2) inhaling aerosol (vaporizing) from THC-predominant cannabis (V-THC), 3) inhaling aerosol from cannabidiol (CBD)-predominant cannabis (V-CBD), or 4) under control conditions. All exercise was performed on a cycle ergometer, with submaximal testing performed at 100 W followed by an evaluation of maximal exercise performance using an all-out 20-min time trial. Metabolism was characterized via the analysis of expired gases while

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subjective ratings of perceived exertion (RPE) were reported. During submaximal cycling, heart rate was higher during S-THC and V-THC compared with both control and V-CBD (all $P < 0.02$). During maximal exercise, $\dot{V}e$ was lower in V-THC compared with control, S-THC, and V-CBD (all $P < 0.03$), as was S-THC compared with control ($P < 0.05$). Both $\dot{V}o2$ and RPE were similar between conditions during maximal exercise (both $P > 0.1$). Mean power output during the 20-min time trial was significantly lower in the S-THC and V-THC conditions compared with both control and V-CBD (all $P < 0.04$). Cannabis containing THC alters the physiological response to maximal and submaximal exercise, largely independent of the inhalation method. THC-containing cannabis negatively impacts vigorous exercise performance during a sustained 20-min effort, likely due to physiological and psychotropic effects. Inhalation of cannabis devoid of THC and primarily containing CBD has little physiological effect on the exercise response or performance.

**New & Noteworthy -** Inhalation of cannabis containing THC alters physiological responses to both submaximal and maximal exercise and reduces mean power output during a 20-min time trial, regardless of whether it is inhaled as smoke or aerosol. In contrast, cannabis devoid of THC and predominantly containing CBD has no effect on physiological responses to exercise or performance.

## *Circadian rhythm in sportspersons and athletic performance: A mini-review. Chronobiology International: The Journal of Biological and Medical Rhythm Research, 41(2): 137-181, 2024.*

**Abstract -** Circadian rhythms in the physiological and behavioral processes of humans play a crucial role in the quality of living and also in the magnitude of success and failure in various endeavors including competitive sports. The rhythmic activities of the body and performance in sportspersons do have a massive impact on their every cutthroat competition. It is essential to schedule sports activities and training of players according to their circadian typology and time of peak performance for improved performance and achievement. In this review, the focus is on circadian rhythms and diurnal variations in peak athletic performance in sportspersons. Accuracy and temporal variability in peak performance in an individual could be attributed to various factors, namely chronotype, time of the day, body temperature, jetlag, hormones, and prior light exposure. Circadian rhythm of mood, alertness, T-core,

and ultimately athletic performance is not only affected by sleep but also by circadian variations in hormones, such as cortisol, testosterone, and melatonin. There are, however, a few reports that are not consistent with the conclusions drawn in this review. Nevertheless, circadian rhythm and performance among sportspersons and athletes are important areas of research. This review might be useful to the managers and policymakers associated with competitive sports and athletic events.

### *Current evidence on common dietary supplements for sleep quality. American Journal of Lifestyle Medicine, Ahead of Print: 1-5, 2024.*

**Abstract -** Poor sleep quality impacts nearly 70 million adults in the United States, resulting in nearly 1 in 5 adults regularly utilizing some sleep aid, either prescription or over-the-counter. Common dietary supplements utilized to improve sleep quality include nitrates, melatonin, magnesium, zinc, vitamin D, and L-theanine. While underlying physiologic mechanisms support the potential impact these compounds have on sleep quality, evidence from clinical trials varies widely. Melatonin and magnesium are two of the more widely researched supplements, which have numerous studies, with conflicting results. Other compounds such as nitrates, zinc, vitamin D, and L-theanine are less supported. Additional research is needed in order to more appropriately recommend these dietary supplements for sleep improvements.

### *Effects of mindfulness-based interventions on promoting athletic performance and related factors among athletes: A systematic review and meta-analysis of randomized controlled trial. International Journal of Environmental Research and Public Health, 20:2038, 2023.*

**Abstract -** In recent years, mindfulness-based interventions (MBIs) have been widely applied in competition sports with respect to athletic performance and mental health promotion, whereas evidence of randomized controlled trials (RCTs) has not been well summarized. Therefore, this study aimed to systematically review and meta-analyze the existing evidence on the effects of MBIs on

improving athletic performance, mindfulness level, mindfulness-related psychological components (e.g., acceptance, self-compassion, flow), and mental health (e.g., burnout, stress, psychological well-being) among athletes. Following the PRISMA guidelines, a literature search was implemented on five electronic databases (Web of Science, PubMed, Scopus, ProQuest, and ScienceDirect) and relevant review papers. The article selection, risk of bias assessment, and data extraction were performed by two investigators independently. The standardized mean difference (SMD) was calculated to evaluate the effects of interventions using the random effect model. Among the 1897 original hits, thirty-two eligible RCT studies were included in the systematic review, of which seven were involved in the meta-analysis. The results showed that MBIs were effective in promoting athletes' athletic performances (by narrative synthesis), mindfulness-level (n = 3; SMD = 0.50, 95% CI = [0.17, 0.83]; I2 = 45%, p = 0.16), and mindfulness-related psychological components (n = 5; SMD = 0.81, 95% CI = [0.53, 1.10], I2 = 77%, p =0.001), while no significant intervention effects were found on the mental health of athletes (n = 4; SMD = -0.03, 95% CI = [-0.35, 0.29], I2 = 89%, p < 0.001). Our findings preliminarily support the potential effectiveness of MBIs, whereas more high-quality RCTs were needed in the future.

## *Exposures to elevated core temperatures during football training: The impact on autonomic nervous system recovery and function. Sports, 12: 8, 2024.*

**Abstract -** Exercising with elevated core temperatures may negatively affect autonomic nervous system (ANS) function. Additionally, longer training duration under higher core temperatures may augment these negative effects. This study evaluated the relationship between exercise training duration and 24 h ANS recovery and function at ≥37 °C, ≥38 °C and ≥39 °C core temperature thresholds in a sample of male Division I (D1) collegiate American football athletes. Fifty athletes were followed over their 25-week season. Using armband monitors (Warfighter MonitorTM, Tiger Tech Solutions, Inc., Miami, FL, USA), core temperature (°C) and 24 h post-exercise baseline heart rate (HR), HR recovery and heart rate variability (HRV) were measured. For HRV, two time-domain indices were measured: the root mean square of the standard deviation of the NN interval (rMSSD) and the standard deviation of the NN interval (SDNN). Linear regression models were performed to evaluate the associations between exercise training duration and ANS recovery (baseline HR and HRV) and function (HR recovery) at ≥37 °C, ≥38 °C and ≥39 °C core temperature thresholds. On average, the athletes were 21.3 (± 1.4) years old,

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weighed 103.0 (±20.2) kg and had a body fat percentage of 15.4% (±7.8%, 3.0% to 36.0%). The duration of training sessions was, on average, 161.1 (±40.6) min and they ranged from 90.1 to 339.6 min. Statistically significant associations between training duration and 24 h ANS recovery and function were observed at both the ≥38.0 °C (baseline HR: $\beta = 0.10 \pm 0.02$, $R2 = 0.26$, $p < 0.0000$; HR recovery: $\beta = -0.06 \pm 0.02$, $R2 = 0.21$, $p = 0.0002$; rMSSD: $\beta = -0.11 \pm 0.02$, $R2 = 0.24$, $p < 0.0000$; and SDNN: $\beta = -0.16 \pm 0.04$, $R2 = 0.22$, $p < 0.0000$) and ≥39.0 °C thresholds ($\beta = 0.39 \pm 0.05$, $R2 = 0.62$, $p < 0.0000$; HR recovery: $\beta = -0.26 \pm 0.04$, $R2 = 0.52$, $p < 0.0000$; rMSSD: $\beta = -0.37 \pm 0.05$, $R2 = 0.58$, $p < 0.0000$; and SDNN: $\beta = -0.67 \pm 0.09$, $R2 = 0.59$, $p < 0.0000$). With increasing core temperatures, increases in slope steepness and strengths of the associations were observed, indicating accelerated ANS deterioration. These findings demonstrate that exercise training under elevated core temperatures (≥38 °C) may negatively influence ANS recovery and function 24 h post exercise and progressively worsen.

## *Football practices in hot environments impact subsequent days' hydration. Journal of Strength and Conditioning Research, 38(1):90-96, 2024.*

**Abstract -** The impact of proper hydration to prevent exertional heat illness in American football has not been evaluated during high school preseason football practices in a hot environment (wet-bulb globe temperature = 31.3 ± 1.8° C). The purposes of this study were to examine the accuracy of urinary hydration measures to assess body mass (BM) changes and to examine carryover effects of consecutive practices by comparing postpractice with the next prepractice values. Before and after each of 7 outdoor practices, 31 male high school football players (age = 16 ± 1 years, height = 181.2 ± 12.0 cm, BM = 85.7 ± 19.1 kg, body mass index = 20.8 ± 1.8) provided a urine sample and were weighed to assess hydration. Sensitivity and specificity of urine color (Ucol) and urine-specific gravity (USG) to determine BM changes were determined using receiver operating characteristic (ROC) analysis. Paired samples t -tests assessed carryover effects between practices. Repeated-measures analysis of variance assessed carryover effects across practices. Significance was set at $p < 0.05$. Sensitivity and specificity for using Ucol or USG to determine BM changes was not significant. For Ucol, there was a carryover effect from practice numbers 2 to 3, 6 to 7 am , 7 am to 7 pm ( $p < 0.001$ for all), and 10 to 11 ( $p = 0.004$); most with less than 24 hours between practices. The %BM loss (%BML) was significantly greater ( $p = 0.001$ to 0.024) after 2-a-day practices. Effects of previous days' exercise in the heat, as evidenced by higher Ucol

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and %BL, are greater after 2-a-day practices, which occurred on later practice days. Athletes must replenish fluids during and between practices to remain euhydrated.

### *Hamstring Injuries: A paradigm for return to play. Seminars in Musculoskeletal Radiology, 28:119-129, 2024.*

**Abstract -** Muscle injuries are the most common sports-related injuries, with hamstring involvement most common in professional athletes. These injuries can lead to significant time lost from play and have a high risk of reinjury. We review the anatomy, mechanisms of injury, diagnostic imaging modalities, and treatment techniques for hamstring injuries. We also present the latest evidence related to return to play (RTP) after hamstring injuries, including a review of articles targeted to RTP in European soccer (Union of European Football Associations), American football (National Football League), and other professional sports. Review of imaging findings in hamstring injury, grading systems for injuries, considerations for RTP, as well as advances in injury prevention, are discussed.

### *Heel Pain – Plantar Fasciitis: Revision 2023. Journal of Orthopaedic and Sports Physical Therapy, 53(12): CPG1-CPG39, 2023.*

**Abstract -** The Orthopaedic Section of the American Physical Therapy Association (APTA) has an ongoing effort to create evidence-based practice guidelines for orthopaedic physical therapy management of patients with musculoskeletal impairments described in the World Health Organization's International Classification of Functioning, Disability, and Health (ICF). The purpose of these revised clinical practice guidelines is to review recent peer-reviewed literature and make recommendations related to nonarthritic heel pain.

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## *Implementing velocity-based training to optimize return to sprint after anterior cruciate ligament reconstruction in soccer players: A clinical commentary. International Journal of Sports Physical Therapy, 19(3):355-365, 2024.*

**Abstract -** After anterior cruciate ligament reconstruction (ACLR), return to sprint is poorly documented in the literature. In soccer, return to sprint is an essential component of return to play and performance after ACLR. The characteristics of running in soccer are specific (velocity differences, nonlinear, intensity). It is important to address these particularities, such as curvilinear running, acceleration, deceleration, changes of direction, and variations in velocity, in the patient's rehabilitation program. Force, velocity, and acceleration capacities are key elements to sprint performance. Velocity-based training (VBT) has gained much interest in recent years and may have a role to play in optimizing return to play and return to sprint after ACLR. Force, velocity, and acceleration can be assessed using force-velocity-power and acceleration-speed profiles, which should inform rehabilitation. The purpose of this commentary is to describe a velocity-based return to sprint program which can be used during ACLR rehabilitation.

## *Initial assessment and management of select musculoskeletal injuries: A team physician consensus statement. Current Sports Medicine Reports, 23(3): 86-104, 2024.*

**Abstract -** Musculoskeletal injuries occur frequently in sport during practice, training, and competition. Injury assessment and management are common responsibilities for the team physician. Initial Assessment and Management of Musculoskeletal Injury-A Team Physician Consensus Statement is title 23 in a series of annual consensus documents written for the practicing team physician. This statement was developed by the Team Physician Consensus Conference, an annual project-based alliance of six major professional associations. The goal of this document is to help the team physician improve the care and treatment of the athlete by understanding the initial assessment and management of selected musculoskeletal injuries.

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***Ketogenic diets are beneficial for athletic performance. Medicine and Science in Sports and Exercise, 56(4): 753-755, 2024. (No abstract available)***

***Ketogenic diets are not beneficial for athletic performance. Medicine and Science in Sports and Exercise, 56(4): 756-759, 2024. (No abstract available)***

## *Low Ankle-GO Score while returning to sport after lateral ankle sprain leads to a 9-fold increased risk of recurrence: A two-year prospective cohort study. Sports Medicine – Open, 10:23, 2024.*

**Background:** Lateral ankle sprain (LAS) is the most common sports injury, leading to a high rate of recurrence and the development of chronic ankle instability. One possible explanation is the lack of objective, evidence-based criteria to inform return to sport decisions following LAS. The aim of this study was therefore to assess the efficacy of a new functional score to distinguish patients at risk of recurrent LAS within two years after the initial injury.

**Methods:** The Ankle-GO score was used in 64 active patients two months after LAS. This composite score includes 2 self-reported questionnaires and 4 functional tests, for a maximum score of 25 points. The rate of reinjury was prospectively recorded 2 years after inclusion. Potential predictive variables for reinjury were tested using the Chi-square and independent t-tests. The area under the receiver operating characteristics curve (AUC) with the optimal cut-off score was determined to assess the predictive value of the Ankle-GO score for the risk of reinjury. Multivariate logistic regression was then used to determine the influence of risk factors of reinjury.

**Results:** Fifty-four (85%) patients were included (23 men and 31 women, 34.7 ± 13 years old) including 18 (33.3%) with a reinjury. The two-month Ankle-GO score was lower in patients with a recurrent LAS (5.4 ± 2.8 points vs. 9.1 ± 4.5, $p = 0.002$) and predicted the risk of reinjury (AUC = 0.75). Patients with < 8 points were found to have a significantly higher risk of reinjury (OR = 8.6; 95%CI: 2-37.2, $p = 0.001$). Women also tend to have a higher risk of recurrence (OR = 3.8; 95%CI: 0.9-15.5, $p = 0.065$).

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**Conclusion:** The Ankle-GO score is a new objective criterion for RTS after LAS. Patients with a low score at two months have a 9-fold greater risk of recurrence within two years.

## *Many pieces to the puzzle: A new holistic workload approach to designing practice in sports. Sports Medicine – Open, 9:38, 2023.*

**Abstract -** Representative learning design (RLD) in sport is a well-established concept in both theory and practice. The goal of RLD is to faithfully replicate competition environments in training settings to benefit improvement in athletic performance. There is currently little research that considers how representative an activity needs to be to facilitate learning transfer, and how that level of representativeness might fluctuate between activities or sessions, and across competitive cycles. Similarly, there is no existing research that specifically considers the elevated workload (in cognitive and physical load) of highly representative training, and the potential impacts of chronic overuse of these highly demanding activities. This paper addresses these limitations, making a case for the application of RLD that considers the level of representativeness (fidelity) and the demands placed on athletes (load) from both a cognitive and physical perspective. This paper also suggests several categorizations of training activities that are based on their relative representativeness, level of imposed demands, and the intended outcomes of the activity with reference to the perception-action cycle. The two core concepts of fidelity and load are combined for a new approach to representative training that allows practitioners to balance the benefits of representative training with the risks of imposing excessive load on athletes.

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# Pro Football Sports Science Research Update

## *Mechanical and metabolic power in accelerated running – Part 2: Team sports. European Journal of Applied Physiology, 124:417-431, 2024.*

**Purpose:** This manuscript is devoted to discuss the interplay between velocity and acceleration in setting metabolic and mechanical power in team sports.

**Methods:** To this aim, an essential step is to assess the individual Acceleration-Speed Profile (ASP) by appropriately analyzing training sessions or matches. This allows one to estimate maximal mechanical and metabolic power, including that for running at constant speed, and hence to determine individual thresholds thereof.

**Results:** Several approaches are described and the results, as obtained from 38 official matches of one team (Italian Serie B, season 2020-2021), are reported and discussed. The number of events in which the external mechanical power exceeded 80% of that estimated from the subject's ASP ([Formula: see text]) was 1.61 times larger than the number of accelerations above 2.5 m s-2 ([Formula: see text]). The difference was largest for midfielders and smallest for attackers (2.30 and 1.36 times, respectively) due to (i) a higher starting velocity for midfielders and (ii) a higher external peak power for attackers in performing [Formula: see text]. From the energetic perspective, the duration and the corresponding metabolic power of high-demanding phases ([Formula: see text]) were essentially constant (6 s and 22 W kg-1, respectively) from the beginning to the end of the match, even if their number decreased from 28 in the first to 21 in the last 15-min period, as a consequence of the increased recovery time between [Formula: see text] from 26 s in the first to 37 s in the last 15-min period.

**Conclusion:** These data underline the flaws of acceleration counting above fixed thresholds.

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## *More than metrics: The role of socio-environmental factors in determining the success of athlete monitoring. Journal of Sports Sciences, Ahead of Print: 1-11, 2024.*

**Abstract -** The perceived value of athlete monitoring systems (AMS) has recently been questioned. Poor perceptions of AMS are important, because where practitioners lack confidence in monitoring their ability to influence programming, and performance is likely diminished. To address this, researchers have primarily sought to improve factors related to monitoring metrics, e.g., validity rather than socio-environmental factors, e.g., buy-in. Seventy-five practitioners (response rate: n = 30) working with Olympic and Paralympic athletes were invited to take part in a survey about their perceptions of AMS value. Fifty-two per cent (n = 13) was confident in the sensitivity of their athlete self-report measures, but only 64% (n = 16), indicated their monitoring was underpinned by scientific evidence. A scientific base was associated with improved athlete feedback ($r_S$ (23) = 0.487, p =0.014*) and feedback correlated with athlete monitoring adherence ($r_S$ (22) = 0.675, p = <0.001**). If athletes did not complete their monitoring, 52% (n = 13) of respondents felt performance might be compromised. However, most respondents 56% (n = 14), had worked with internationally successful athlete(s) who did not complete their monitoring. While AMS can be a useful tool to aid performance optimization, its potential value is not always realized. Addressing socio-environmental factors alongside metric-factors may improve AMS efficacy.

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## *NCAA football players are at higher risk of upper extremity injury after first-time concussion. The Physician and Sportsmedicine, Ahead of Print: 1-16, 2024.*

**Background:** Previous research has demonstrated that concussions increase the risk of subsequent lower extremity musculoskeletal injury in athletes. However, the risk of upper extremity injury in athletes' post-concussion is poorly understood.

**Methods:** All concussed football players within a National Collegiate Athletic Association (NCAA) Division I conference athletic database were identified between 2017 and 2021. After exclusions, each athlete experiencing their first concussion was then retrospectively reviewed for upper extremity injuries in the year prior to their concussion and in the year beginning at 90 days after their concussion. All upper extremity injuries were identified and the odds ratio, 95% confidence interval, and statistical significance between groups were calculated in Microsoft Excel.

**Results:** 160 de-identified football players from a single conference who were first diagnosed with concussions in the seasons from 2017 through 2021 met inclusion criteria. In these athletes the odds of upper extremity injury in year following first diagnosed concussion were 2.36 times higher than in the year prior (95% CI 1.13-4.95, $p = 0.02$). Shoulder was the most common site of injury with 57.7% of injuries compared to 19.2% in the hand, 15.4% in the elbow, 7.7% in the forearm, and 0% in the wrist.

**Conclusion:** This study demonstrates that collegiate football players are at a 2.36 times greater risk of upper extremity injury in the year following their first diagnosed concussion compared to the year preceding it. The most common site of upper extremity injury after concussion was the shoulder.

**Level of evidence:** III.

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# Pro Football Sports Science Research Update

## *On the value of a premium college football player: Evaluating the literature. Journal of Sports Economics, Ahead of Print: 1-35, 2024.*

**Abstract -** Over the past decade, the issue of player compensation in college sports has been the subject of several successful legal challenges. Athletes contend that the compensation they receive falls significantly short of the value they generate, attributing this gap to unlawful National Collegiate Athletic Association restrictions. Numerous tools exist in the sports economic literature that estimate the value of college athletes, with an emphasized focus toward premium college football players. In addition to providing updated estimate of player marginal revenue product (MRP), we review past and contemporary methodologies for estimating college player MRPs. We contend that, while presenting some evidence that restrictions on player compensation resulted in the extraction of the majority of the value generated by top college athletes, existing methods leave considerable uncertainty over the magnitude of exploitation.

## *Patellar tendon load progression during rehabilitation exercises: Implications for the treatment of patellar tendon injuries. Medicine and Science in Sports and Exercise, 56(3):545-552, 2024.*

**Purpose:** This study aimed to evaluate patellar tendon loading profiles (loading index, based on loading peak, loading impulse, and loading rate) of rehabilitation exercises to develop clinical guidelines to incrementally increase the rate and magnitude of patellar tendon loading during rehabilitation.

**Methods:** Twenty healthy adults (10 females/10 males, 25.9 ± 5.7 yr) performed 35 rehabilitation exercises, including different variations of squats, lunge, jumps, hops, landings, running, and sports specific tasks. Kinematic and kinetic data were collected, and a patellar tendon loading index was determined for each exercise using a weighted sum of loading peak, loading rate, and cumulative loading impulse. Then the exercises were ranked, according to the loading index, into tier 1 (loading index ≤0.33), tier 2 (0.33 < loading index <0.66), and tier 3 (loading index ≥0.66).

**Results:** The single-leg decline squat showed the highest loading index (0.747). Other tier 3 exercises included single-leg forward hop (0.666), single-leg countermovement jump (0.711), and running cut

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(0.725). The Spanish squat was categorized as a tier 2 exercise (0.563), as was running (0.612), double-leg countermovement jump (0.610), single-leg drop vertical jump (0.599), single-leg full squat (0.580), double-leg drop vertical jump (0.563), lunge (0.471), double-leg full squat (0.428), single-leg 60° squat (0.411), and Bulgarian squat (0.406). Tier 1 exercises included 20 cm step up (0.187), 20 cm step down (0.288), 30 cm step up (0.321), and double-leg 60° squat (0.224).

**Conclusions:** Three patellar tendon loading tiers were established based on a combination of loading peak, loading impulse, and loading rate. Clinicians may use these loading tiers as a guide to progressively increase patellar tendon loading during the rehabilitation of patients with patellar tendon disorders and after anterior cruciate ligament reconstruction using the bone-patellar tendon-bone graft.

## *Performance-enhancing substances in sport: A scientometric review of 75 years of research. Drug Testing and Analysis, Ahead of Print: 1-12, 2024.*

**Abstract -** The use of performance-enhancing substances not only undermines the core values of sports but also poses significant health risks to athletes. In a fast-evolving doping environment, where sport professionals are constantly seeking novel and illegal means to bypass doping tests, and new substances are regularly detected on the drug market, it is crucial to inform authorities with updated evidence emerging from scientific research. The current study aims to (i) outline the structure of knowledge in the literature on performance enhancers in sports (i.e., most active countries, main sources, most productive authors, and most frequently used keywords); (ii) identify the most impactful documents in the field; and (iii) uncover the main domains of research in the literature. To do so, we conducted a comprehensive scientometric analysis of the literature on doping, sourcing our data from Scopus. Our research involved a document co-citation analysis of 193,076 references, leading to the identification of the 51 most influential documents and seven key thematic areas within the doping literature. Our results indicate that the scientific community has extensively studied the most prevalent doping classes, such as anabolic agents and peptide hormones, and little is still known about the use of contaminated supplements or other types of enhancers identified as emergent trends. Concurrently, technological advancements contributed to the development of more sophisticated doping detection techniques, using blood or urine samples. More recently, the focus has shifted towards the athlete biological passport, with research efforts aimed at identifying biomarkers indicative of doping. The dynamic nature

of doping methods underlines the necessity for more robust educational campaigns, aiming at raising awareness among sports professionals and their entourage about the dangers of doping and the intricacies of its control mechanisms.

### *A profile of isometric cervical strength in elite professional male rugby players. Journal of Orthopaedic and Sports Physical Therapy, 54(3):1-8, 2024.*

**Objectives:** To (1) provide position-specific normative data for isometric cervical muscle strength and endurance in professional, male rugby players and (2) assess the relationship between age, height, weight, and playing position with cervical muscle strength and endurance.

**Design:** Cross-sectional study.

**Methods:** Professional rugby players completed peak isometric cervical strength testing followed by a test of cervical muscle endurance. Descriptive statistics for continuous variables with strength normalized to body weight were performed. Multiple linear regression was used to estimate associations between strength measurements.

**Results:** In total, 136 players participated including front-row forwards (27%), other forwards (28%), and backs (45%). Front-row forwards had significantly greater peak isometric cervical muscle strength than other position groups, with backs having the lowest strength. Extension produced the highest force for all 3 position groups (429 N ± 104 N), whereas flexion produced the least (275 N ± 65 N). Age was associated with increased isometric cervical muscle strength. There was a statistically significant relationship between peak flexion strength and flexion endurance (P = .003). The average time for the endurance tests were 55.7 (±17.1) seconds and 52.9 (±20.1) seconds for extension and flexion, respectively. Other forwards had lower cervical extension muscle endurance than backs and front-row forwards.

**Conclusions:** Normative values for peak and endurance isometric strength in professional rugby players illustrate significant differences between playing position. Consider age, body weight, and intraindividual variability when interpreting cervical strength and endurance results.

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## *Recovery management in sport: Overview and outcomes of a nine-year multicenter research program. International Journal of Sports Science & Coaching, Ahead of Print: 1-11, 2024.*

**Abstract -** The nine-year multicenter research project, titled REGman—Optimization of Training and Competition: Management of Regeneration in Elite Sports—aimed to advance the understanding of recovery management in sports and generate practical recommendations. In this article, we outline the overarching research program that links the specific research strands and outputs during the two federal funding cycles (2012–2021) and summarize the main project findings. The two-stage conceptual framework involved investigating monitoring measures for short-term fatigue and recovery, as well as assessing the efficacy of various recovery interventions. These interventions encompassed psychological relaxation and recovery strategies, sleep, cooling techniques like precooling, percooling, cold water immersion, or whole-body cryotherapy, compression garments, active recovery, stretching, heating interventions such as sauna or contrast water therapy, massage, and foam rolling. The findings revealed inconclusive or marginal effects of recovery interventions at the group level, while indicating possible interindividual differences in responses. Additionally, the findings highlighted the effectiveness of diverse monitoring measures, showing satisfactory sensitivity in tracking performance changes related to fatigue and recovery. The use of individualized reference ranges significantly improved classification accuracy compared to group-based reference ranges. Athletes and coaches are encouraged to prioritize fundamental aspects of training and recovery: meticulous training planning and execution, effective sleep management, and proper nutrition. Furthermore, monitoring and analyzing individual responses, even though it demands suitable methodologies and presents challenges in high-performance sports environments, can yield valuable insights for personalized recovery management. If these aspects are comprehensively addressed, and resources allow, additional recovery strategies might be explored.

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# Pro Football Sports Science Research Update

## *Repetitive head injuries in German American football players do not change blood-based biomarker candidates for CTE during a single season. Neurological Research and Practice, 5:13, 2024.*

**Background:** Repetitive traumatic brain injuries in American football players (AFPs) can lead to the neurodegenerative disease chronic traumatic encephalopathy (CTE). Clinical symptoms of CTE range from mood and behavioral changes to cognitive impairment, depression, and suicidality. So far, CTE cannot be diagnosed in vivo and thus specific diagnostic parameters for CTE need to be found, to observe and treat exposed athletes as early as possible. Promising blood-based biomarkers for CTE include total tau (tTau), hyperphosphorylated tau (pTau), neurofilament light protein (NF-L), glial fibrillary acidic protein (GFAP), amyloid-β40 (Aβ40), amyloid-β42 (Aβ42) and calcium-binding protein B (S100-B). Previous studies have found elevated levels of these biomarkers in subjects exposed to TBIs, whereas cerebrospinal fluid (CSF) levels of Aβ40 and Aβ42 were decreased in CTE subjects. Here, we investigated whether young AFPs already exhibit changes of these biomarker candidates during the course of a single active season.

**Methods:** Blood samples were drawn from n = 18 American Football Players before and after a full season and n = 18 male age-matched control subjects. The plasma titers of tTau, pTau, NF-L, GFAP, Aβ40, Aβ42 and S100-B were determined. Additionally, Apathy, Depression, and Health status as well as the concussion history and medical care were assessed and analyzed for correlations.

**Results:** Here we show, that the selected biomarker candidates for CTE do not change significantly during the seven-month period of a single active season of American Football in blood samples of AFPs compared to healthy controls. But interestingly, they exhibit generally elevated pTau titers. Furthermore, we found correlations of depression, quality-of-life, career length, training participation and training continuation with headache after concussion with various titers.

**Conclusion:** Our data indicates, that changes of CTE marker candidates either occur slowly over several active seasons of American Football or are exclusively found in CSF. Nevertheless, our results underline the importance of a long-term assessment of these biomarker candidates, which might be possible through repeated blood biomarker monitoring in exposed athletes in the future.

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***The 11 evidence-informed and inferred principles of microcycle periodization in elite football. Sport Performance & Science Reports, Feb: 218, 2024. (No abstract available)***

***The differentiation of the competitive athlete with physiologic cardiac remodeling from the athlete with cardiomyopathy. Current Problems in Cardiology, Ahead of Print: 1-47, 2024.***

**Abstract -** There are currently 5 million active high school, collegiate, professional, and master athletes in the United States. Regular intense exercise by these athletes can promote structural, electrical and functional remodeling of the heart, which is termed the "athlete's heart." In addition, regular intense exercise can lead to pathological adaptions that promote or worsen cardiac disease. Many of the athletes in the United States seek medical care. Consequently, physicians must be aware of the normal cardiac anatomy and physiology of the athlete, the differentiation of the normal athlete heart from the athlete with cardiomyopathy, and the contemporary care of the athlete with a cardiomyopathy. In athletes with persistent cardiovascular symptoms, investigations should include a detailed history and physical examination, an ECG, a transthoracic echocardiogram, and in athletes in whom the diagnosis is uncertain, a maximal exercise stress test or a continuous ECG recording, and cardiac magnetic resonance imaging or a cardiac computed tomography angiography when definition of the coronary anatomy or characterization of the aorta and the aortic great vessels is indicated. This article discusses the differentiation of the normal athlete with physiologic cardiac remodeling from the athlete with hypertrophic, dilated or arrhythmogenic ventricular cardiomyopathy (ACM). The ECG changes in trained athletes that are considered normal, borderline, or abnormal are listed. In addition, the normal echocardiographic measurements for athletes who consistently participate in endurance, power, combined or heterogeneous sports are enumerated and discussed. Algorithms are listed that are useful in the diagnosis of trained athletes with borderline or abnormal echocardiographic measurements suggestive of cardiomyopathies along with the major and minor criteria for the diagnosis of ACM in athletes. Thereafter, the treatment of athletes with hypertrophic, dilated, and arrhythmogenic right ventricular cardiomyopathies are reviewed. The distinction between physiologic changes and pathologic

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changes in the hearts of athletes has important therapeutic and prognostic implications. Failure by the physician to correctly diagnose an athlete with hypertrophic cardiomyopathy, dilated cardiomyopathy, or ACM, can lead to the sudden cardiac arrest and death of the athlete during training or sports competition. In contrast, an incorrect diagnosis by a physician of cardiac pathology in a normal athlete can lead to an unnecessary restriction of athlete training and competition with resultant significant emotional, psychological, financial, and long-term health consequences in the athlete.

## *The impact of different warm-up strategies on acceleration and deceleration demands in highly trained soccer players. European Journal Sport Science, 24:88-96, 2024.*

**Abstract -** This study compared the differences in acceleration and deceleration demands between three different warm-up (WU) strategies (Reaction speed [exercises with reaction to a stimulus], Run [self-paced running], and Speed [exercises such as shuttle running or circuits]) applied to highly trained soccer players. Nineteen players were monitored for 4 weeks using a 10 Hz Global Positioning System. Accelerations and decelerations magnitudes were classified as low (25%–50%), moderate (50%–75%), and high (>75%) intensities. Additionally, efforts were analyzed according to their respective starting speeds (<5, 5–10, 10–15, 15–20, 20–25, and >25 km h−1). Differences between WU strategies were estimated via paired mean differences along with effect sizes. The three WU strategies led to few efforts starting >15 km h−1 and high-intensity efforts (<1 effort per minute). Players performed more high-intensity accelerations during Speed than Reaction Speed (ES: 0.74 [90% CI: 0.21, 1.33]); more moderate-intensity accelerations during Reaction Speed than Run (ES: 1.29 [90% CI: 0.72, 2.00]); more moderate-intensity decelerations during Reaction Speed than Run (ES: 0.64 [90% CI: 0.04, 1.32]) and Speed (ES: 0.89 [90% CI: 0.37, 1.50]); more decelerations started at 20–25 km h−1 during Speed than Reaction Speed (ES: 0.81 [90% CI: 0.20, 1.49]) and Run (ES: 0.76 [90% CI: 0.42, 1.18]); and more decelerations started at >25 km h−1 during Speed than Reaction Speed (ES: 3.57 [90% CI: 2.88, 4.58]). In conclusion, Speed elicited higher acceleration and deceleration demands than the Reaction Speed and Run WU strategies, and this should be considered when designing training sessions.

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**Highlights**

- Soccer warm-ups (WUs) use different training strategies, which should be monitored independently as they elicit different acceleration and deceleration demands.
- Speed exercises (such as shuttle running and circuits) may expose players to more high-speed displacements and high-intensity accelerations.
- Practitioners should choose WUs strategies considering the subsequent activity to avoid under or overloading players.

## *The mental health of elite-level coaches: A systematic scoping review. Sports Medicine – Open, 10:16, 2024.*

**Background:** Elite-level coaches are exposed to multiple performance, organizational and personal stressors which may contribute to reduced mental health and wellbeing. This systematic scoping review examined the current body of evidence to explore what is known about the mental health of elite-level coaches (i.e. wellbeing and mental ill-health), the risk and protective factors that influence coach mental health, and the relationship between mental health and coaching effectiveness.

Methods: The review adhered to the Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) guidelines. A systematic search was undertaken and updated in September 2022 using six electronic databases.

**Results:** 12,376 studies were identified and screened, with 42 studies satisfying the inclusion criteria. Despite the paucity of high-quality research, findings indicated that 40% of the included studies examined themes connected to wellbeing, with 76% assessing the nature or prevalence of mental ill-health in elite-level coaches. Among studies exploring mental ill-health, coach burnout was the primary focus, while scant research examined symptoms associated with clinical disorders (e.g. anxiety and depression). Overall, psychological outcomes for elite-level coaches were shaped by risk and protective factors operating at the individual, interpersonal, organizational and societal level. Preliminary evidence was also found to suggest that poor mental health may contribute to reduced coaching effectiveness. It

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is proposed that coaching effectiveness could therefore be employed as a 'hook' to engage elite-level coaches in greater consideration of their mental health needs.

**Conclusion:** Alongside the development of methodologically robust research, there is a need to examine dynamic individual (e.g. psychological skills), interpersonal (e.g. strong social supports) and organizational (e.g. workload) factors that aim to preserve the mental health and optimize the efficacy of elite-level coaches.

## *The sleep parameters of Olympic athletes: Characteristics and assessment instruments. International Journal of Sports Medicine, Ahead of Print: 1-9, 2024.*

**Abstract -** This systematic review aims to identify the sleep parameters of Olympic athletes and the instruments used to assess and monitor the sleep of these athletes. The search was conducted until February 2023 and was performed in PubMed, Web of Science, and Scopus databases. This systematic review has included studies that investigated at least one of the following sleep parameters: total sleep time (TST), sleep onset latency (SOL), sleep efficiency (SE), awakenings after sleep onset (WASO), quality of sleep, daytime sleepiness, and chronotype; the participants were Olympic athletes. The search returned a total of 280 studies. After screening based on exclusion and inclusion criteria, 11 studies were included. The main results demonstrate that Olympic athletes have TST of 06:10 h, SE of 84%, SOL of 28 min, and WASO of 49 min. The most predominant chronotype is indifferent; over half of the athletes have poor sleep quality and complaints. Furthermore, actigraphy was the most used method to assess sleep. It is concluded that Olympic athletes have TST, SE, and WASO poor than the recommended values. In addition, sleep complaints and poor sleep quality were also observed. Among the objective sleep assessment methods, actigraphy was the method most frequently used in this population.

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## *The top 100 most impactful articles on the Achilles tendon according to altmetric attention score and number of citations. The Orthopaedic Journal of Sports Medicine, 12(3): 1-5, 2024.*

**Background:** Achilles tendon injuries often generate substantial discussion in the mainstream media. The Altmetric Attention Score (AAS) measures the online dialogue pertaining to Achilles tendon research that occurs outside scientific journals, which traditional citation-based metrics fail to capture.

**Purpose:** To characterize the top 100 most-cited Achilles tendon articles and compare them with the 100 Achilles tendon articles with the highest AAS. A secondary goal was to gain an improved understanding of the online dissemination and interpretation of Achilles tendon research through this comparison.

**Study design:** Cross-sectional study.

**Methods:** The Web of Science Clarivate database was queried to isolate the 100 most-cited Achilles tendon articles, and the Altmetric database was queried to identify the Achilles tendon articles with the top 100 AAS values. Data elements were extracted for each article including study type, study topic, and geographic origin.

**Results:** The Web of Science Clarivate database search yielded 10,890 articles published between 1970 and 2021. The 100 most-cited articles were published in 35 journals, with the American Journal of Sports Medicine being the most prevalent. The mean (±SD) number of citations was 214.5 ± 86.47. The most prevalent study type was laboratory (28.0%). The most prevalent study topic was treatment (41.0%). Of these articles, 72.0% were European. The Altmetric database search yielded 3810 articles published between 1957 and 2021. The AAS of the top 100 articles ranged from 37 to 476 with a mean of 98.17 ± 85.53. The selected articles were published in 39 journals, with the British Journal of Sports Medicine being the most prevalent. The most prevalent study type was randomized controlled trial (25.0%). The most common study topic was treatment (40.0%). Of these articles, 46.0% were European.

**Conclusion:** Our findings suggest that, although the scientific community remains committed to high-impact journals with articles backed by high citation numbers, there is an increasing opportunity to consume Achilles tendon literature through social media.

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## *Validity of global positioning system technology to measure maximum velocity sprinting in elite sprinters. Journal of Strength & Conditioning, 37(12):2438-2442, 2023.*

**Abstract -** The objective of this study was to assess the concurrent validity of 10-Hz wearable Global Positioning System (GPS) technology to measure maximum velocity sprinting (Vmax) relative to Doppler radar in elite sprinters. Data were collected from a single training session performed by elite 100 and 200 m sprinters (males: n = 5; 100 m best times: 10.02 ± 0.07 seconds, range: 9.94-10.10 seconds; 200 m best times: 20.29 ± 0.42 seconds, range: 19.85-20.80 seconds; females: n = 2; age: 28.0 ± 4.2 years; body mass: 65.8 ± 4.6 kg; 100 m best times: 11.18 ± 0.34 seconds; 200 m best times: 22.53 ± 0.04 seconds). Velocity and time data from 16 maximal, 60-m sprint efforts were recorded simultaneously with 10 Hz GPS and 47 Hz radar. Validity was assessed using Bland-Altman 95% limits of agreement (LOA) and intraclass correlation coefficient (ICC), each with respective 95% confidence intervals (CI). Vmax measured with 10 Hz GPS demonstrated a LOA of -0.11 m·s-1 (-0.17, -0.05) and an ICC of 0.99 (0.98, 1.0) relative to the radar device.10 Hz GPS overestimated Vmax by 0.11 m·s-1 relative to the radar but could still be considered a suitable tool for monitoring external load in elite sprinters. However, the much smaller average annual improvement in this population (~0.1-0.2%) in comparison with the ~1% overestimation reduces the utility of 10 Hz GPS to detect meaningful performance changes in maximum velocity.

## *Wearable and ingestible technology to evaluate and prevent exertional heat illness: A narrative review. PM & R: The Journal of Injury, Function, and Rehabilitation, Ahead of Print: 1-6, 2024.*

**Abstract -** Exertional heat illness remains a constant threat to the athlete, military service member, and laborer. Recent increases in the number and intensity of environmental heat waves places these populations at an ever-increasing risk and can be deadly if not recognized and treated rapidly. For this reason, it is extremely important for medical providers to guide athletes, service members, and laborers in the implementation of awareness, education, and measures to reduce or mitigate the risk of exertional heat illness. Within the past 2 decades, a variety of wearable technology options have

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become commercially available to track an estimation of core temperature, yet questions continue to emerge as to its use, effectiveness, and practicality in athletics, the military, and the workforce. There is a paucity of data on the accuracy of many of these newer devices in the setting of true heat stroke physiology, and it is important to avoid overreliance on new wearable technology. Further research and improvement of this technology are critical to identify accuracy in the diagnosis and prevention of EHI.

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